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Effect of cultural practices on soil properties, moisture conservation and grain yield of winter sorghum (*Sorghum bicolar* L. Moench) in semi-arid tropics of India

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Abstract

A field experiment was laid out during winter seasons of 1994–1995 and 1995–1996 on deep black clayey soils (Vertisols) at Regional Research Station, Bijapur, in the northern dry zone of Karnataka State (Zone 3) of south India to evaluate the effect of cultural practices on soil moisture conservation, soil properties, root growth and yield of sorghum (*Sorghum bicolar* L. Moench). Lay out of plots with in situ moisture conservation practices reduced bulk density, increased infiltration rate, porosity, improved root growth and grain yield of winter sorghum. Conservation and availability of higher amount of moisture and nutrients during various stages of crop growth with moisture conservation practices resulted in better crop growth with higher amount of dry matter production and its translocation to ear in winter sorghum. Compartmental bunding and ridges and furrows increased the grain yield by 22.8 and 25.6% (mean of 1994–1995 and 1995–1996), respectively, over flat bed with similar trend observed during 1994–1995 and 1995–1996. Among organic sources, incorporation of *Leucaena* loppings improved soil physico-chemical properties, conserved higher amount of moisture and increased winter sorghum yield to a greater extent than farmyard manure and vermicompost. Average grain yield (1994–1995 and 1995–1996) of winter sorghum increased by 11.7% with *Leucaena* application as compared to vermicompost. Grain yield increased significantly by 20% with application of 25 kg N ha^{-1} and further increase in nitrogen dose up to 50 kg ha^{-1}, increased the grain yield by 30.5% in the pooled data.
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Keywords: In situ moisture conservation practices; Organic sources; Nitrogen; Black soil; Sorghum

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1. Introduction

Due to various socio-economic-politico-demographic reasons, the mismanagement and misuse of the natural resources have intensified in recent times, leading to degradation, denudation and destruction of natural resources and with proper management of natural resources, viz., land, water and vegetation can reverse the above process. Interest in vertisols in northern dry zone of Karnataka (semi-arid tropics of South India) arises from the predominance of these soils in the major cropping areas. These soils are not only thirsty but are also hungry due to low and erratic rainfall (<650 mm annually). These soils have low infiltration rate (<10 mm h^{-1}) and are undulating in topography with average land slope varying from 1 to 2% on the cultivated lands that causes high runoff (15–20% of annual rainfall). In addition, these soils are low in organic carbon, available nitrogen and low to medium in available phosphorus and medium to high in available potassium.

The above constraints really make the farming in these regions unstable with lower crop yields. The improvement of crop yields in these regions can only be achieved through proper land and water management practices that improve the physico-chemical properties in vertisols, reduce erosion and runoff and facilitate the safe disposal of excess water. Adoption of in situ moisture conservation practices, viz., compartmental bunding (CB) and ridges and furrows (RF) reduces the runoff, causing the water to infiltrate and be stored in the profile, so that it is made available to the crop during various stages of crop growth and especially in the moisture stress situations resulting in better crop growth with higher crop yields (Mahale et al., 1998). Walton (1962) observed higher moisture retention in the soil profile under tied ridges over flat cultivation and concluded that when planting is delayed under a bi-modal rainfall system, tied ridging will enhance crop yields in Uganda.

In recent times, with the adoption of improved crop cultivars that respond to moisture and nutrients, there is a need for application of higher quantities of chemical fertilizers to maintain the soil fertility in this region. The poor socio-economic conditions of farmers, increased price of fertilizers and low and ill distribution of rainfall in this region really raises the risks to farmers in application of higher quantities of fertilizers. The use of locally available plant and animal residues as a part of substitute for nutrients, not only increase the use efficiency of applied fertilizers but also improves the physico-chemical properties of soil, thereby improving the crop yields on sustainable basis. Sorghum is the staple food grown in this region, which meets the food, fodder and fuel requirements. The yields of sorghum in this region during the post-rainy season (*rabi*) are not only unstable but also low due to insufficient moisture and nutrients for normal growth during cropping season.

An attempt was therefore, made to investigate the effect of cultural practices on soil physical properties, moisture conservation and grain yield of winter sorghum in the dryland situations of northern Karnataka State in India.

2. Materials and methods

2.1. Soil and site characteristics

A field experiment was conducted during winter seasons (*rabi*) of 1994–1995 and 1995–1996 on vertisols (deep black clayey soils) at Regional Research Station, Bijapur, which

Table 1
General characteristics of experimental site

Properties	Value	Method		
Physical properties				
Mechanical analysis		International pipette method, Piper (1966)		
Coarse sand (%)	6.3			
Fine sand (%)	18.5			
Silt (%)	14.9			
Clay (%)	60.3			
Infiltration rate (cm h^{-1})	0.7	Double ring infiltrometer method, Black (1965)		
Bulk density (Mg m^{-3})		Core sampler method, Black (1965)		
0–9 cm	1.20			
9–18 cm	1.27			
18–27 cm	1.30			
	Soil depth (cm)			
	0–15	15–30	30–45	
Chemical properties				
Soil pH	8.5	8.7	8.7	pH meter, Piper (1966)
Electrical conductivity (dS m^{-1})	0.27	0.30	0.32	Conductivity bridge
Organic carbon (g kg^{-1})	3.9	3.6	3.3	Wet oxidation method, Piper (1966)
Available nitrogen (kg ha^{-1})	128	120	112	Alkaline permanganate method, Subbiah and Asija (1959)
Available phosphorus (P_2O_5) (kg ha^{-1})	28	26	20	Olsen's method, Jackson (1967)
Available potassium (kg ha^{-1})	412	396	360	Flame photometer, Muhr et al. (1965)

is situated in northern dry zone of Karnataka State, India, at $16^0 49^1$N latitude, $75^0 42^1$E longitude and at an altitude of 593.8 m above mean sea level. The physical and chemical properties of soil (prior to start of experiment) are presented in Table 1.

2.2. Treatments

A field experiment was laid out on lands having 1.0% slope in split–split plot design with three replications. Moisture conservation treatments, viz., CB and RF were imposed during third week of July in 1994 and third week of June during 1995 to the main plot. Compartmental *bunds* of size 3 m × 3 m and ridges and furrows at 60 cm were formed with bund former and by bullock drawn ridger, respectively. In the subplot, *Leucaena* loppings (at 2.5 t ha^{-1}) and farmyard manure (at 2.5 t ha^{-1}) were incorporated during third week of August and first week of September, respectively. Vermicompost (at 1.0 t ha^{-1}) was applied manually at the time of sowing. In the sub-subplot, nitrogen fertilizer (0, 25 and 50 kg ha^{-1}) as per treatments through urea and recommended dose of phosphorus (25 kg ha^{-1}) through single super phosphate were applied before sowing. Maladandi 'M35-1', a *rabi* (post-rainy season) sorghum cultivar was sown on 4 October 1994 and 14 September 1995 to a depth of 5 cm at 15 cm apart in rows of 60 cm and harvested on 15 February 1995 and 25 January 1996, respectively. Grain and straw yield from net plot was harvested, sun dried, weighed and further converted to kg ha^{-1}/q ha^{-1}. A root study was undertaken in the experimental

plots at the end of second season (1995–1996), after harvest of sorghum crop. Soil profile in the experimental plots were recharged and made completely wet, a day prior to the root studies. With the help of a small pitcher of water from tap (water source), water was sprayed to the roots from the surface and roots were removed and measured for their length, spread and were oven dried to record weight.

Soil properties were determined at the start of the experiment and at harvest after every year of the experimentation by employing standard procedures described by Jackson (1967). Infiltration rate was recorded after the harvest of crop during 1994–1995 and 1995–1996 in first replication using double ring infiltrometer having a height of 30 cm and diameter of 20 and 40 cm for the inner and outer cylinders, respectively, as described by Richards (1954). Bulk density was measured in undisturbed soil cores collected from 0 to 0.09, 0.09–0.18 and 0.18–0.27 m soil depths after harvest of crop during both the years of study (Black, 1965). From bulk density, total porosity was calculated assuming a particle density of 2.65 Mg m^{-3}. Profile soil water was gravimetrically determined at 0–15, 15–30 and 30–60 cm soil depths in each treatment at sowing, 30, 60, 90 days after sowing (DAS) and at harvest of the crop. Soil moisture utilized was computed as the difference in soil moisture at sowing, 30, 60, 90 DAS and at harvest.

3. Results

3.1. Soil properties

3.1.1. In situ moisture conservation practices

In the present study, CB and RF recorded higher profile water content in top 0.60 m soil depth as compared to flat bed (FB) at sowing, various stages of crop growth and at harvest during both years of experimentation (Fig. 1). Bulk density over the seasons did not change, however, there was reduction in bulk density during the second year of study in all the soil depths. Formation of compartmental *bunds* and RF increased the infiltration rate to the higher extent over FB during both the years of study with higher magnitude during second year (1995–1996) (Tables 2 and 3). At the end of second year of experimentation (1995–1996), bulk density in compartmental bunded and RF formed plots was lower (1.22 and 1.18 Mg m^{-3}, respectively) as compared to FB (1.25 Mg m^{-3}) in 0–0.09 m soil depth. Similar trend was followed in 0.09–0.18 and 0.18–0.27 m soil depths. The values for bulk density in 0.09–0.18 m soil depths for FB, CB and RF were 1.30, 1.25 and 1.21 Mg m^{-3}, respectively, and the corresponding values for 0.18–0.27 m soil depths were 1.33, 1.29 and 1.23, respectively (Table 3). Similar trend as that of the above was also followed at the end of the first year of experimentation (1994–1995) with little higher values in all soil depths (Table 2). As expected, topsoil (0–0.09 m) recorded higher porosity than subsoil depths (0.09–0.18 and 0.18–0.27 m). At the end of second season (1995–1996), the porosity increased from 53.0 (FB) to 53.9% (CB) and 55.5% (RF) in 0–0.09 m soil depth and the corresponding values of porosity during 1994–1995 were 52.2, 52.9 and 53.6% for FB, CB and RF, respectively (Table 4). The porosity values decreased in subsoil depths during both the years of study. There was little change in organic carbon content in flat bed as compared to the original soil (Tables 1 and 5). The

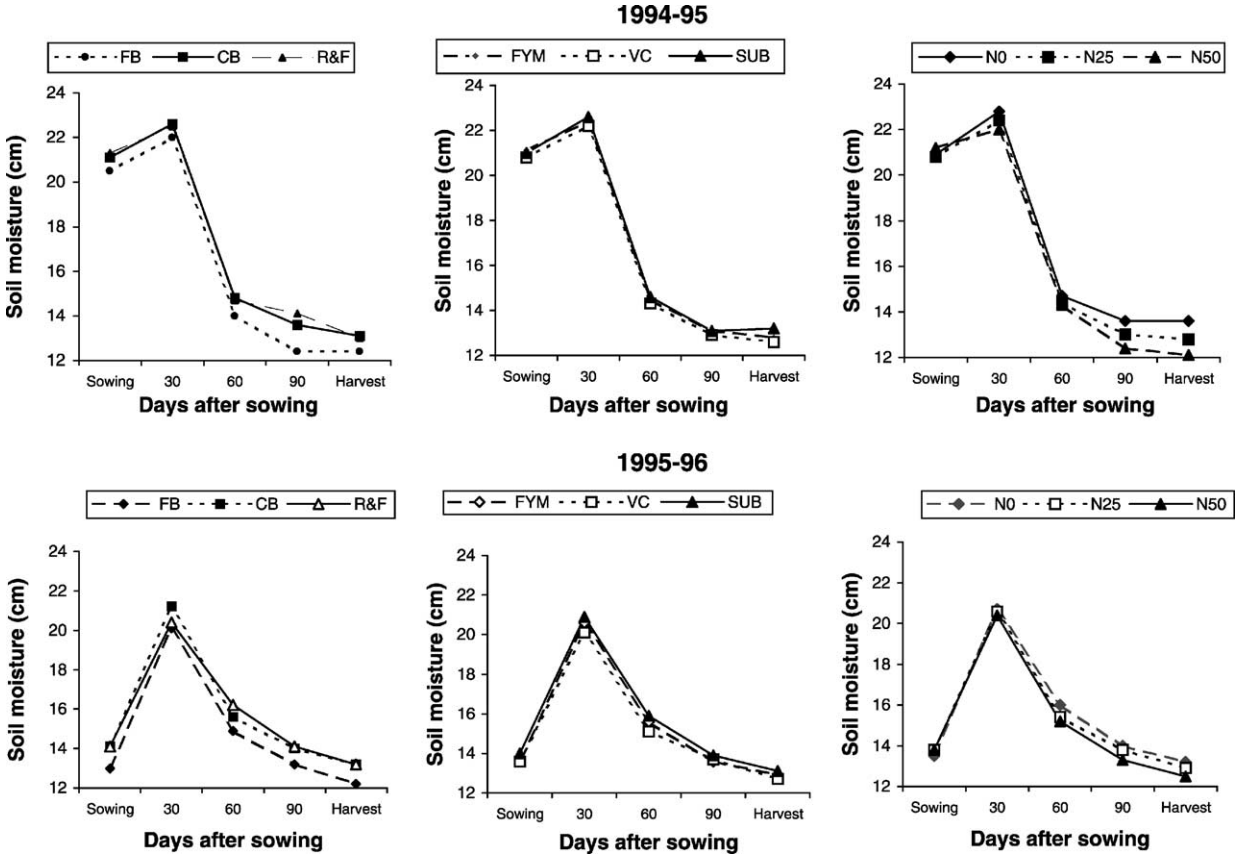


Fig. 1. Soil moisture in top 0.60 m soil profile as influenced by in situ moisture conservation practices, organic sources and nitrogen levels at various stages of crop growth during winter seasons (1994–1995 and 1995–1996).

Table 2
Infiltration rate and bulk density at harvest (1994–1995) as influenced by in situ moisture conservation practices, organic sources and nitrogen levels

Treatments		Infiltration rate (cm h^{-1})				Bulk density (Mg m^{-3}) in different soil depths											
In situ[a]	Organic sources[a]	0 kg ha^{-1}	25 kg ha^{-1}	50 kg ha^{-1}	Mean	0–9 cm				9–18 cm				18–27 cm			
						0 kg ha^{-1}	25 kg ha^{-1}	50 kg ha^{-1}	Mean	0 kg ha^{-1}	25 kg ha^{-1}	50 kg ha^{-1}	Mean	0 kg ha^{-1}	25 kg ha^{-1}	50 kg ha^{-1}	Mean
FB	FYM at 2.5 t ha^{-1}	0.67	0.63	0.71	0.67	1.25	1.27	1.25	1.26	1.28	1.36	1.29	1.31	1.30	1.37	1.31	1.33
	VC at 1.0 t ha^{-1}	0.62	0.63	0.60	0.62	1.27	1.29	1.29	1.28	1.28	1.29	1.38	1.32	1.30	1.31	1.40	1.34
	LEU at 2.5 t ha^{-1}	0.67	0.70	0.67	0.68	1.25	1.28	1.27	1.26	1.28	1.36	1.31	1.32	1.29	1.38	1.32	1.33
	Mean	0.65	0.66	0.66	0.66	1.25	1.28	1.27	1.27	1.28	1.34	1.33	1.31	1.30	1.35	1.34	1.33
CB	FYM at 2.5 t ha^{-1}	0.81	0.85	0.86	0.84	1.25	1.26	1.23	1.25	1.26	1.34	1.30	1.30	1.27	1.34	1.31	1.31
	VC at 1.0 t ha^{-1}	0.79	0.81	0.80	0.80	1.24	1.26	1.25	1.25	1.28	1.31	1.34	1.31	1.30	1.32	1.36	1.32
	LEU at 2.5 t ha^{-1}	0.90	0.81	0.87	0.86	1.27	1.17	1.31	1.25	1.28	1.26	1.314	1.29	1.30	1.28	1.31	1.29
	Mean	0.84	0.82	0.84	0.83	1.25	1.23	1.26	1.25	1.28	1.30	1.32	1.30	1.29	1.31	1.33	1.31
RF	FYM at 2.5 t ha^{-1}	0.87	0.86	0.93	0.88	1.20	1.22	1.26	1.23	1.23	1.25	1.26	1.25	1.24	1.24	1.28	1.25
	VC at 1.0 t ha^{-1}	0.84	0.88	0.83	0.85	1.18	1.28	1.24	1.23	1.25	1.26	1.28	1.26	1.29	1.27	1.28	1.28
	LEU at 2.5 t ha^{-1}	0.94	0.94	0.91	0.93	1.20	1.26	1.24	1.23	1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.25
	Mean	0.89	0.89	0.89	0.89	1.20	1.25	1.25	1.23	1.24	1.25	1.26	1.25	1.26	1.26	1.27	1.26
Mean	FYM at 2.5 t ha^{-1}	0.78	0.78	0.83	0.80	1.23	1.25	1.25	1.24	1.26	1.32	1.28	1.29	1.27	1.32	1.30	1.30
	VC at 1.0 t ha^{-1}	0.75	0.78	0.74	0.56	1.23	1.28	1.26	1.26	1.27	1.29	1.33	1.30	1.30	1.30	1.34	1.31
	LEU at 2.5 t ha^{-1}	0.84	0.82	0.82	0.82	1.24	1.23	1.27	1.25	1.27	1.29	1.29	1.28	1.28	1.30	1.29	1.29
	Mean	0.79	0.79	0.80	0.79	1.23	1.25	1.26	1.25	1.27	1.30	1.30	1.29	1.28	1.31	1.31	1.29

[a] FB: flat bed; CB: compartmental bunding; RF: ridges and furrows; FYM: farmyard manure; VC: vermicompost; LEU: *Leucaena* loppings.

Table 3
Infiltration rate and bulk density at harvest (1995–1996) as influenced by in situ moisture conservation practices, organic sources and nitrogen levels

Treatments		Infiltration rate (cm h^{-1})				Bulk density (Mg m^{-3}) in different soil depths											
In situ[a]	Organic sources[a]	0 kg ha^{-1}	25 kg ha^{-1}	50 kg ha^{-1}	Mean	0–9 cm				9–18 cm				18–27 cm			
						0 kg ha^{-1}	25 kg ha^{-1}	50 kg ha^{-1}	Mean	0 kg ha^{-1}	25 kg ha^{-1}	50 kg ha^{-1}	Mean	0 kg ha^{-1}	25 kg ha^{-1}	50 kg ha^{-1}	Mean
FB	FYM at 2.5 t ha^{-1}	0.70	0.74	0.78	0.74	1.22	1.25	1.25	1.24	1.27	1.30	1.30	1.29	1.28	1.31	1.32	1.30
	VC at 1.0 t ha^{-1}	0.71	0.76	0.66	0.71	1.23	1.28	1.26	1.26	1.31	1.30	1.28	1.30	1.33	1.34	1.30	1.32
	LEU at 2.5 t ha^{-1}	0.75	0.77	0.80	0.77	1.23	1.25	1.25	1.24	1.35	1.36	1.25	1.32	1.37	1.38	1.32	1.36
	Mean	0.72	0.76	0.75	0.74	1.23	1.26	1.25	1.25	1.31	1.32	1.28	1.30	1.33	1.34	1.31	1.33
CB	FYM at 2.5 t ha^{-1}	0.91	0.94	0.98	0.95	1.23	1.26	1.19	1.23	1.24	1.26	1.25	1.25	1.25	1.31	1.32	1.30
	VC at 1.0 t ha^{-1}	0.90	0.94	0.96	0.90	1.25	1.23	1.23	1.23	1.28	1.27	1.28	1.23	1.28	1.31	1.31	1.30
	LEU at 2.5 t ha^{-1}	0.95	0.96	0.99	0.97	1.19	1.20	1.22	1.20	1.22	1.23	1.24	1.23	1.28	1.25	1.29	1.27
	Mean	0.92	0.95	0.94	0.94	1.22	1.23	1.21	1.22	1.24	1.26	1.26	1.25	1.27	1.29	1.31	1.29
RF	FYM at 2.5 t ha^{-1}	0.95	0.99	0.99	0.97	1.20	1.15	1.17	1.17	1.20	1.20	1.22	1.21	1.22	1.21	1.24	1.22
	VC at 1.0 t ha^{-1}	0.94	0.92	0.92	0.93	1.17	1.19	1.24	1.20	1.21	1.22	1.27	1.23	1.23	1.26	1.30	1.26
	LEU at 2.5 t ha^{-1}	0.97	1.01	1.02	1.00	1.14	1.16	1.18	1.16	1.18	1.19	1.19	1.19	1.20	1.21	1.22	1.21
	Mean	0.95	0.97	0.98	0.97	1.17	1.17	1.19	1.18	1.20	1.20	1.22	1.21	1.22	1.23	1.25	1.23
Mean	FYM at 2.5 t ha^{-1}	0.85	0.89	0.92	0.89	1.22	1.22	1.20	1.21	1.24	1.25	1.26	1.25	1.25	1.28	1.29	1.27
	VC at 1.0 t ha^{-1}	0.85	0.87	0.81	0.85	1.22	1.23	1.24	1.23	1.27	1.27	1.27	1.27	1.28	1.30	1.30	1.29
	LEU at 2.5 t ha^{-1}	0.89	0.92	0.94	0.91	1.19	1.20	1.22	1.20	1.25	1.26	1.23	1.25	1.28	1.28	1.28	1.28
	Mean	0.86	0.89	0.89	0.88	1.21	1.22	1.22	1.22	1.25	1.26	1.25	1.26	1.27	1.29	1.29	1.28

[a] FB: flat bed; CB: compartmental bunding; RF: ridges and furrows; FYM: farmyard manure; VC: vermicompost; LEU: *Leucaena* loppings.

Table 4
Porosity (%) in different soil depths during both the years of study as influenced by in situ moisture conservation practices, organic sources and nitrogen levels

Treatments	1994–1995			1995–1996		
	0–0.09 m	0.09–0.18 m	0.18–0.27 m	0–0.09 m	0.09–0.18 m	0.18–0.27 m
In situ moisture conservation practices						
Flat bed	52.2	50.5	49.8	53.0	50.8	49.9
Compartmental bunding	52.9	51.0	50.6	53.9	52.7	51.4
Ridges and furrows	53.6	52.7	52.4	55.5	54.4	53.5
Organic sources						
Farmyard manure at 2.5 t ha^{-1}	53.1	51.5	51.1	54.2	52.8	52.0
Vermicompost at 1.0 t ha^{-1}	52.6	51.1	50.5	53.6	52.2	51.2
Leucaena loppings at 2.5 t ha^{-1}	52.9	51.5	51.3	54.6	53.0	51.7
Nitrogen levels (kg ha^{-1})						
0	53.4	52.2	51.7	54.4	52.8	52.0
25	52.8	51.0	50.7	54.0	52.5	51.5
50	52.5	50.9	50.5	54.0	52.7	51.3

organic carbon content increased with formation of CB (4.3 g kg^{-1}) and RF (4.4 g kg^{-1}) as compared to FB (3.9 g kg^{-1}) in 0–0.15 m soil depth and the similar trend was also followed in 0.15–0.30 m depth with lower values at end of second year of experimentation (Table 5).

Table 5
Organic carbon content (g kg^{-1}) in different soil depths at harvest of second season (1995–1996) as influenced by in situ moisture conservation practices, organic sources and nitrogen levels

Treatments	Soil depths	
	0–15 cm	15–30 cm
In situ moisture conservation practices		
Flat bed	3.9	3.6
Compartmental bunding	4.3	4.0
Ridges and furrows	4.4	4.1
Organic sources		
Farmyard manure at 2.5 t ha^{-1}	4.2	4.0
Vermicompost at 1.0 t ha^{-1}	3.9	3.5
Leucaena loppings at 2.5 t ha^{-1}	4.5	4.2
Nitrogen levels (kg ha^{-1})		
0	4.0	3.5
25	4.2	3.9
50	4.4	4.3

3.1.2. Organic sources

Leucaena and farmyard manure incorporation in higher quantity (at 2.5 t ha^{-1}) over application of lower quantity of vermicompost at 1.0 t ha^{-1} recorded slightly higher soil moisture in top 0.60 m soil profile from sowing till harvest during both the years of study (Fig. 1). *Leucaena* incorporated plots recorded higher infiltration rate of 0.91 cm h^{-1} (7%) as compared to vermicompost (0.85 cm h^{-1}) during 1995–1996. Similar trend was observed during 1994–1995 with lower values. During 1995–1996, incorporation of *Leucaena* loppings recorded lower bulk density of 1.21, 1.25 and 1.28 Mg m^{-3} as compared to vermicompost application (1.23, 1.27 and 1.29 Mg m^{-3}) in 0–0.09, 0.09–0.18 and 0.18–0.27 m soil depths, respectively (Table 3). The values of bulk density were lower during 1995–1996 as compared to 1994–1995 with incorporation of different organic sources in all the soil depths. Application of *Leucaena* also recorded higher porosity in top (54.6%) and subsoil depths (53.0 and 51.7%) and vermicompost application resulted in lower porosity (53.6, 52.2 and 51.2%, respectively) in all the soil depths (0–0.09, 0.09–0.18 and 0.18–0.27 m, respectively) during 1995–1996 (Table 4). The corresponding values for different organic sources and with increase in soil depths were lower during 1994–1995 as compared to 1995–1996. Mean organic carbon content was higher (4.2 g kg^{-1}) in topsoil as compared to subsoil (3.9 g kg^{-1}). Incorporation of *Leucaena* recorded higher organic carbon of 4.5 and 4.2 g kg^{-1} as compared to vermicompost (3.9 and 3.5 g kg^{-1}) in topsoil and subsoil depths, respectively (Table 5).

3.1.3. Nitrogen levels

Reduction in soil moisture in top 0.60 m soil profile from 30 DAS until harvest with increase in nitrogen fertilizer up to 50 kg ha^{-1} was observed during both the years of study (Fig. 1). With increase in nitrogen fertilizer up to 50 kg ha^{-1} reduced the infiltration rate and increased the bulk density. Higher values of bulk density were observed in subsoil depths as compared to topsoil depth (0–0.09 m) during both the years of study (Tables 2 and 3). Increase in N fertilizer (0–50 kg ha^{-1}) reduced porosity in all depths of soil during both the years of study with maximum porosity of 54.4% was observed in 0–0.09 m soil depth at harvest (1995–1996) in plots that received no N fertilizer (Table 4). Application of 50 kg N ha^{-1} recorded maximum values of organic carbon (4.4 and 4.3 g kg^{-1}) in topsoil and subsoil, respectively, over lower doses of nitrogen (Table 5).

3.1.4. Interaction

At harvest of winter sorghum during 1995–1996 lay out of plots with RF with *Leucaena* and 50 kg N ha^{-1} application recorded higher infiltration rate (1.02 cm h^{-1}) (Table 3). Minimum bulk density (1.14 Mg m^{-3}) with maximum porosity (56.8%) was observed in plots laid out with RF with *Leucaena* application and without nitrogen addition in topsoil (0–0.09 m). Similar trend in bulk density and porosity was also observed in subsoil depths, with slightly higher bulk density (1.25 and 1.28 Mg m^{-3}) and lower porosity (53 and 51.8%). Flat bed with vermicompost and 50 kg N ha^{-1} application resulted in lower infiltration rate, higher bulk density and lower porosity in both topsoil and subsoil depths.

3.2. Root studies

3.2.1. In situ moisture conservation practices

Compartmental *bund*s and RF recorded appreciable increase in root depth (48.1 and 51.9 cm, respectively) over FB (39.4 cm) (Table 6). Maximum root spread of 53.4 cm was observed in RF followed by CB (46.7 cm) and varied significantly over FB (34.8 cm). Ridges and furrows and CB recorded higher root weight of 12.58 and 11.74 g per plant, respectively, and varied significantly over FB (7.95 g per plant).

3.2.2. Organic sources

Incorporation of *Leucaena* resulted in maximum root depth (48.4 cm) as compared to farmyard manure (45.7 cm) and vermicompost (45.3 cm) (Table 6). Even, maximum root spread of 47.0 cm around sorghum plant was observed with *Leucaena* incorporation and it varied significantly over vermicompost (41.4 cm). *Leucaena* incorporation thus resulted in higher root biomass of 11.68 g per plant followed by vermicompost (10.37 g per plant).

3.2.3. Nitrogen levels

Increase in nitrogen dose up to 25 kg ha^{-1} increased the root depth (47.3 cm), root spread (45.2 cm) and root dry weight (10.72 g per plant) over control and further increase in nitrogen dose up to 50 kg ha^{-1} improved the root growth marginally, with maximum root depth (48.8 cm), root spread (46.2 cm) and root weight (11.66 g per plant) (Table 6).

3.2.4. Interaction

Application of *Leucaena* and 25 kg N ha^{-1} in plots laid out with RF recorded the maximum root depth (57.2 cm), root spread (58.7 cm), whereas root weight was higher with the application of 50 kg N ha^{-1} (15.23 g per plant). Flat bed with vermicompost and without nitrogen application recorded minimum root depth (34.7 cm) and root spread (29.0 cm). The lowest root weight of 5.10 g per plant was observed in FB with farmyard manure and without nitrogen application (Table 6).

3.3. Grain yield and straw yield

3.3.1. In situ moisture conservation practices

During 1994–1995, increase in grain yield with CB was by 26.8% (1570 kg ha^{-1}) as compared to 18.9% increase (1563 kg ha^{-1}) during 1995–1996 over FB (1238 and 1314 kg ha^{-1}, respectively). Further, with adoption of RF, grain yield increased to the higher extent of 33.9 (1658 kg ha^{-1}) and 17.7% (1547 kg ha^{-1}) during 1994–1995 and 1995–1996 over FB, respectively. In pooled analysis, grain yield increased by 22.8 and 25.6% with CB and RF over FB (Table 7 and Fig. 2). The trend in straw yield was also similar to that of grain yield during both the years of study and in the pooled analysis.

3.3.2. Organic sources

There was not much variation in winter sorghum grain yield with application of either farmyard manure or *Leucaena*. *Leucaena* incorporation recorded significantly higher grain yield of 1582 (1994–1995), 1557 (1995–1996) and 1570 kg ha^{-1} (pooled) over lower grain

Table 6
Root length, root spread and root weight of winter sorghum as influenced by in situ moisture conservation practices, organic sources and nitrogen levels during 1995–1996 at harvest

Treatments		Root length (cm)				Root spread (cm)				Root weight (g per plant)			
In situ[a]	Organics[a]	0 kg ha^{-1}	25 kg ha^{-1}	50 kg ha^{-1}	Mean	0 kg ha^{-1}	25 kg ha^{-1}	50 kg ha^{-1}	Mean	0 kg ha^{-1}	25 kg ha^{-1}	50 kg ha^{-1}	Mean
FB	FYM at 2.5 t ha^{-1}	35.5	41.0	42.0	39.5	30.8	39.4	36.1	35.4	5.10	8.20	9.00	7.43
	VC at 1.0 t ha^{-1}	34.7	35.9	38.6	36.4	29.0	29.3	35.3	31.2	7.53	6.93	8.20	7.56
	LEU at 2.5 t ha^{-1}	37.9	43.9	45.4	42.4	35.9	37.2	40.2	37.8	7.93	8.27	10.37	8.86
	Mean	36.0	40.3	42.0	39.4	31.9	35.3	37.2	34.8	6.86	7.80	9.19	7.95
CB	FYM at 2.5 t ha^{-1}	46.8	46.6	52.9	48.8	45.6	46.4	50.7	47.6	9.60	11.37	11.70	10.89
	VC at 1.0 t ha^{-1}	42.4	48.0	46.2	45.5	38.8	44.6	45.0	42.8	11.17	11.57	12.00	11.58
	LEU at 2.5 t ha^{-1}	45.3	52.1	52.9	50.1	44.3	52.2	53.0	49.8	12.03	12.90	13.30	12.74
	Mean	44.8	48.9	50.9	48.1	42.9	47.7	49.6	46.7	10.93	11.94	12.33	11.74
RF	FYM at 2.5 t ha^{-1}	46.7	47.3	52.9	48.9	46.9	47.2	48.4	47.5	11.37	13.33	12.30	12.33
	VC at 1.0 t ha^{-1}	53.7	54.1	54.2	54.0	46.2	52.1	53.0	47.5	12.37	10.70	12.87	11.98
	LEU at 2.5 t ha^{-1}	47.2	57.2	53.8	52.7	46.0	58.7	55.4	50.2	11.83	13.23	15.23	13.43
	Mean	49.2	52.9	53.6	51.9	46.4	52.7	52.0	53.4	11.86	12.42	13.47	12.58
Mean	FYM at 2.5 t ha^{-1}	43.0	45.0	49.3	45.7	41.1	44.3	45.1	43.5	8.69	10.97	11.00	10.22
	VC at 1.0 t ha^{-1}	43.6	46.0	46.0	45.3	38.0	42.0	44.2	41.4	10.36	9.73	11.02	10.37
	LEU at 2.5 t ha^{-1}	43.5	51.1	50.7	48.4	42.1	49.4	49.5	47.0	10.60	11.47	12.97	11.68
	Mean	43.4	47.3	48.8	46.5	40.4	45.2	46.2	44.0	9.88	10.72	11.66	10.76

[a] FB: flat bed; CB: compartmental bunding; RF: ridges and furrows; FYM: farmyard manure; VC: vermicompost; LEU: *Leucaena* loppings.

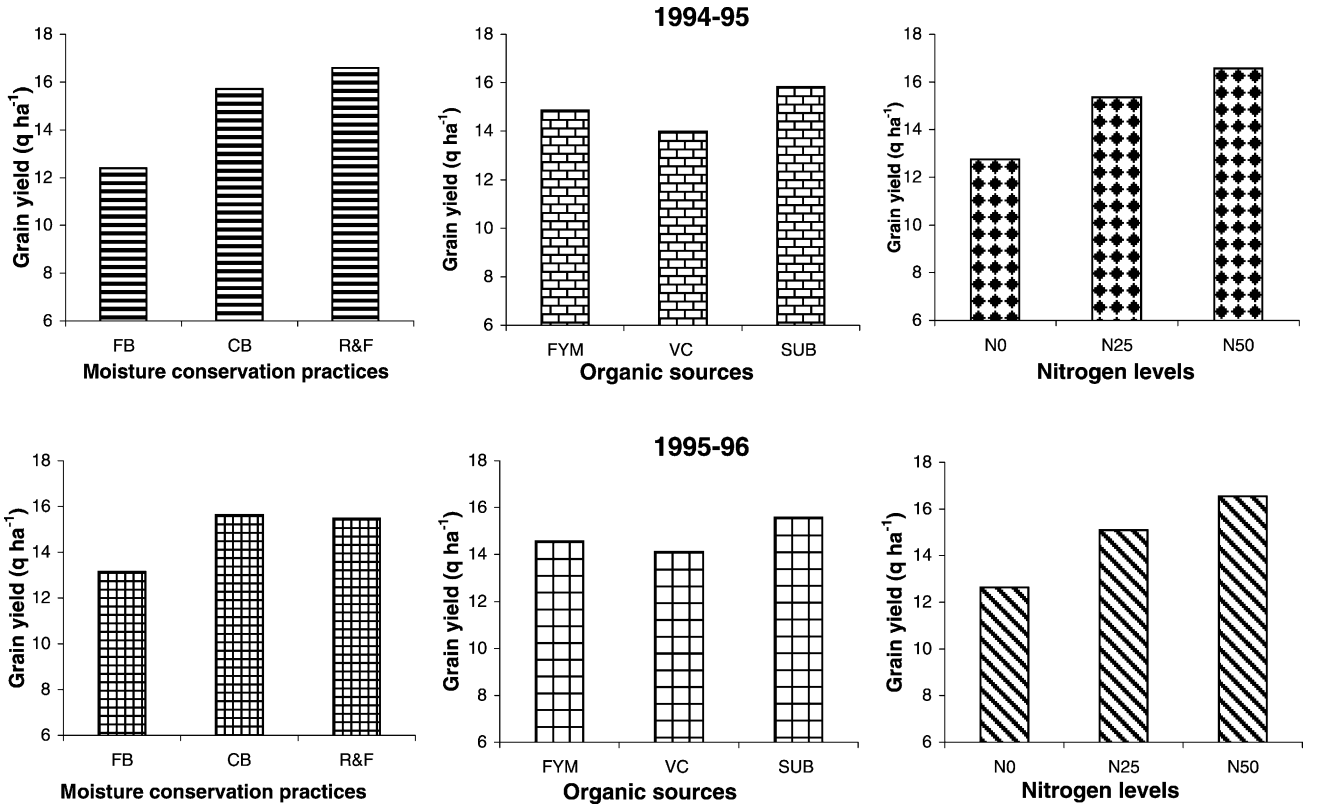


Fig. 2. Effect of moisture conservation practices, organic sources and nitrogen on grain yield of winter sorghum during 1994–1995 and 1995–1996.

Table 7
Grain and straw yield of winter sorghum as influenced by in situ moisture conservation practices, organic sources and nitrogen levels

Treatments	Grain yield (kg ha^{-1})			Straw yield (kg ha^{-1})		
	1994–1995	1995–1996	Pooled	1994–1995	1995–1996	Pooled
In situ moisture conservation practices						
Flat bed	1238	1314	1276	1499	1637	1568
Compartmental bunding	1570	1563	1567	1855	1993	1924
Ridges and furrows	1658	1547	1603	1999	1902	1950
S.Em±	60	44	38	66	50	41
CD ($P = 0.05$)	246	172	125	259	195	135
Organic sources						
Farmyard manure at 2.5 t ha^{-1}	1486	1457	1405	1796	1850	1823
Vermicompost at 1.0 t ha^{-1}	1398	1411	1472	1687	1782	1735
Leucaena loppings at 2.5 t ha^{-1}	1582	1557	1570	1870	1900	1885
S.Em±	51	36	31	34	43	28
CD ($P = 0.05$)	157	112	91	106	NS[a]	81
Nitrogen levels (kg ha^{-1})						
0	1274	1263	1268	1552	1588	1570
25	1536	1509	1522	1854	1869	1862
50	1657	1653	1655	1947	2075	2011
S.Em±	38	53	25	36	38	26
CD ($P = 0.05$)	108	95	69	104	107	72

[a] Non-significant.

yield observed in vermicompost during both the years of study and in the pooled data (Table 7 and Fig. 2). *Leucaena* application recorded 11.7% higher yield over vermicompost (pooled). The trend in straw yield was similar as that of grain yield with *Leucaena* application recording 8.6% higher (1885 kg ha^{-1}) as compared to vermicompost (1735 kg ha^{-1}).

3.3.3. Nitrogen levels

Grain and straw yield increased significantly with application of nitrogen up to 50 kg ha^{-1} during 1994–1995, 1995–1996 and in the pooled data over lower doses. Grain yield increased by 20 (1522 kg ha^{-1}) and 30.5% (1655 kg ha^{-1}) with application of 25 and 50 kg N ha^{-1}, respectively, in the pooled data (Table 7 and Fig. 2). In the pooled data, straw yield increased by 18.6 and 28.1% with application of 25 and 50 kg N ha^{-1} over control.

4. Discussion

At inter-terrace level, adopting in situ moisture conservation practices can control soil and water losses and improve the soil moisture, nutrient status and sustain the yields of winter sorghum in the vertisols of northern dry region of Karnataka State, India. In the present investigation, grain yield increased by 26.8 and 18.9% with CB and 33.9 and 17.7% with RF as compared to FB during 1994–1995 and 1995–1996, respectively. In pooled analysis, grain yield increased by 22.8 and 25.6% with CB and RF as compared to FB (Table 7 and

Fig. 2). The trend in straw yield was also similar to that of grain yield during both the years of study and in the pooled analysis. The magnitude of increase in yield with adoption of in situ moisture conservation practices was higher during first year of study (1994–1995), though crop-experienced drought at physiological maturity as compared to the second year of study (1995–1996), when the crop was sown early with uniform distribution of rainfall and without moisture stress occurrence during cropping season (Table 8). This clearly proves that the adoption of in situ moisture conservation practices was more beneficial under drought year with higher amount of moisture conservation as compared to normal year. Adoption of in situ moisture conservation practices conserved the rain water and increased the grain yields of different crops especially during drought years in different locations of India in vertisols (Robinson et al., 1986; Kandiannan et al., 1992; More et al., 1996; Velayutham et al., 1997; Patil, 2001, 2002).

Higher winter sorghum yields in CB and RF was mainly due to higher profile water content in top 0.60 m soil depth with in situ moisture conservation practices as compared to FB at sowing and at various stages of crop growth and at harvest, during both years of study (Fig. 1) (Fairbourn and Gardner, 1974; Randhawa and Rama Mohan Rao, 1981; Mittal et al., 1986; Radder et al., 1991; Surkod, 1993; Mastiholi, 1994). Higher soil moisture in top 0.60 m soil profile in plots laid out with CB and RF was due to formation of miniature *bund*s/barriers that restricted the flow of rainwater and increased the time of concentration. This has ultimately reduced runoff and soil loss, increased infiltration rate and porosity. The increase in porosity with adoption of moisture conservation practices was due to decrease in bulk density in different soil depths. On the contrary, porosity decreased with increase in soil depth and was due to compaction and increase in bulk density (Tables 2–4).

The organic carbon content increased with formation of CB (4.3 g kg^{-1}) and RF (4.4 g kg^{-1}) as compared to FB (3.9 g kg^{-1}) in 0–0.15 m soil depth and the similar trend was also followed in 0.15–0.30 m depth with lower values at the end of second year of study (Table 5). Higher organic carbon content with moisture conservation practices over FB was attributed to greater root density and higher amount of crop residues added through root biomass. Feller et al. (1987) earlier reported that the roots are also main source of crop residues. It was observed that the roots decompose more rapidly than the equivalent amount of residue left at the soil surface (Table 5).

The development of root system in spread, depth and density depends mainly on the quantum and rate of water and nutrient uptake by plants particularly under dryland conditions and especially during drought years. The increase in the values of the root weight, spread and depth with adoption of in situ moisture conservation practices was attributed to improved soil physical properties and increased moisture conditions (Tables 2–4 and Fig. 1). This indicates that the adoption of in situ moisture conservation practices provided optimum conditions for root growth. Similar to above findings, increase in rooting depths and root density for agricultural crops using subsoiler for seedbed preparation have been earlier reported by several investigators (Bennie and Botha, 1986; Ross, 1986; Barbosa et al., 1989; Nitant and Singh, 1995; Bhan et al., 1995; Singh and Verma, 1996).

Organic sources incorporation increases the porosity and infiltration rate, reduces the runoff and soil loss, thus conserving the soil moisture. This in turn increases the use efficiency of the fertilizers applied under dryland conditions. There was not much variation in winter sorghum yield with application of either farmyard manure or *Leucaena*. *Leucaena*

Table 8
Meteorological data for the years 1994–1995 and 1995–1996 and the average meteorological data of 56 years (1939–1994)

Month	Normal weather parameters (1939–1994)					Rain fall (mm)		Number of rainy days		Maximum temperature (°C)		Minimum temperature (°C)		Relative humidity (%)	
	Rainfall (mm)	Number of rainy days	Maximum temperature (°C)	Minimum temperature (°C)	Relative humidity (%)	1994–1995	1995–1996	1994–1995	1995–1996	1994–1995	1995–1996	1994–1995	1995–1996	1994–1995	1995–1996
April	20.1	1.5	38.0	24.0	51	30.6	20.9	3	3	36.8	38.7	24.5	21.7	55	53
May	48.1	4.0	38.4	24.0	59	9.3	45.6	1	5	39.7	37.3	24.9	21.0	52	60
June	100.9	8.4	33.3	22.5	75	64.9	21.5	5	3	31.5	36.4	22.8	21.5	77	59
July	81.3	9.9	30.2	21.8	80	20.8	48.9	3	7	29.7	30.4	21.8	20.2	80	64
August	88.7	7.8	30.1	21.3	80	26.5	67.3	4	6	30.5	30.8	22.1	19.6	77	66
September	156.7	10.4	30.6	21.1	80	6.6	238.4	1	10	31.9	30.4	21.8	18.9	74	64
October	105.5	6.3	31.0	20.6	71	339.5	175.0	12	12	30.7	29.7	23.5	19.5	78	66
November	29.1	2.8	29.7	17.5	61	8.6	11.8	3	1	28.2	30.3	17.6	15.4	77	56
December	7.9	1.1	28.9	15.4	59	0.0	0.0	0	0	28.3	29.9	16.9	17.1	70	52
January	2.3	0.6	30.2	16.1	57	76.4	0.0	3	0	29.0	31.4	13.6	18.1	58	55
February	2.4	0.2	32.9	17.9	48	0.0	0.0	0	0	30.8	35.1	15.7	19.6	52	46
March	6.9	1.0	36.0	21.3	46	2.6	0.0	1	0	35.5	37.9	19.1	22.4	51	37
Total	649.9	54.0	–	–	–	585.8	629.4	36	47	–	–	–	–	–	–

incorporation recorded significantly higher grain and straw yield by 11.7 and 8.6% (pooled), respectively, over vermicompost (Table 7 and Fig. 2). Higher yields with application of *Leucaena* or farmyard manure over vermicompost was due to increased soil moisture content in the top 0.60 m profile to the higher extent at different stages of crop growth during both the years of study with similar results being earlier observed in the above soils (Bellakki and Badanur, 1994). Increased moisture content in soil profile with *Leucaena* application was due to higher fine and coarse aggregates with increased porosity, mean weight diameter and hydraulic conductivity and this might have increased infiltration rate and reduced bulk density (Badanur et al., 1990; Badanur and Malabasari, 1995; Ghuman et al., 1997; Mishra and Sharma, 1997) (Tables 2–4). Higher yields with *Leucaena* loppings over farmyard manure and vermicompost was also due to higher availability of organic carbon. Higher organic carbon in topsoil was mainly due to application of *Leucaena* and other organic sources in topsoil. Higher organic carbon with incorporation of *Leucaena* over the rest of the organics was mainly due to higher rate of mineralization and faster decomposition with increased nutrient availability (Bellakki and Badanur, 1994; Venkateswarlu, 1984; Ramamoorthy et al., 2002).

Application of *Leucaena* not only improved the soil physico-chemical properties but also increased the use efficiency of the fertilizers applied. This in turn resulted in better root development (root depth, spread and volume) with higher winter sorghum yield. Similar findings in root length of rice, wheat and winter maize were observed by Mishra and Sharma (1997), wherein, root length enhanced with farmyard manure + blue green algae (BGA) application compared to no manure, farmyard manure alone, BGA alone in the silty loam soils of Bihar.

The vertisols of semi-arid tropics of India are low in available nitrogen and hence, addition of nitrogen plays an important role in the nutrition of winter sorghum crop grown especially in the drylands of northern Karnataka. Conservation of moisture through in situ moisture conservation practices during rainy season and addition of increased doses of nitrogen interacted positively and increased the crop yields. Grain and straw yield increased significantly with application of nitrogen up to 50 kg ha^{-1} during 1994–1995, 1995–1996 and in the pooled data as compared to 25 kg N ha^{-1} and control. Grain and straw yield increased by 20 and 18.6% and 30.5 and 28.1%, respectively, with application of 25 and 50 kg N ha^{-1} in the pooled data (Table 7 and Fig. 2). Increase in crop yield with increase in N fertilizer up to 50 kg ha^{-1} was due to increased availability of nutrients especially the nitrogen. This in turn improved the root growth and increased the N uptake further resulted in higher rate of photosynthesis, dry matter production and its translocation to ear as indicated by increased grain and straw yield. Studies conducted by Surkod (1993) at Bijapur, Durgude et al. (1996) at Sholapur and Rama Mohan Rao et al. (1995), Patil (2001, 2002) at Bellary, in vertisols of dryland region of South India, indicated increased yields of sorghum with increase in nitrogen application up to 60 kg ha^{-1}, depending upon rainfall situation during the cropping season. Moisture and nitrogen interactions are additive to increase the crop yields at higher doses of nitrogen under good years of rainfall in dryland situations. Reduction in soil moisture in top 0.60 m soil profile from 30 DAS until harvest with increase in nitrogen fertilizer up to 50 kg ha^{-1} was due to utilization of moisture by the plants in production of higher dry matter (Figs. 1 and 2). Lower infiltration rate, porosity and higher bulk density with increased N fertilizer up to 50 kg ha^{-1} may be attributed to deterioration

in soil structure due to dilution of soil matrix with the loss of loose material (organic carbon) as well as due to deterioration in aggregate stability (Bellakki, 1995) (Tables 2–4). Application of 50 kg N ha^{-1} recorded maximum values of organic carbon (4.4 and 4.3 g kg^{-1}) in top and subsoil, respectively, over lower doses of nitrogen (Table 5). Higher organic carbon content with increase in N dose up to 50 kg ha^{-1} was due to improved root growth that resulted in higher root biomass which was more accumulated in topsoil due to higher root spread as compared to subsoil. The higher root biomass decomposed during the course of time to produce higher organic carbon content.

5. Conclusion

The present studies results clearly indicate that in situ moisture conservation practices, viz., CB and RF in the vertisols of semi-arid tropics of South India, not only increases in conservation of moisture and nutrients but also helps in increasing their availability. The conservation of moisture is more useful during drought years as compared to normal and above normal years of rainfall in Indian dryland situations. The winter sorghum yield increased by 22.8 and 25.6% with adoption of CB and RF, respectively, as compared to FB (pooled). The higher yield was mainly due to improved soil physico-chemical properties, viz., increased infiltration rate, porosity and higher moisture in top 0.60 m soil profile with improved root growth. Application of *Leucaena* at 2.5 t ha^{-1} proved beneficial in improving soil properties, with improved biomass production of both underground (roots) and aboveground (shoot) parts and finally the grain yield by 11.7% (pooled) over application of vermicompost at 1.0 t ha^{-1}. Increase in N fertilizer up to 50 kg ha^{-1}, increased the grain yield significantly as compared to application of 0, 25 kg ha^{-1} during 1994–1995 and 1995–1996. In the pooled data, application of 25 and 50 kg N ha^{-1} increased the grain yield significantly by 20 and 30.5%, respectively, over control. Conservation of moisture in situ during rainy season with integrated nutrient management improves the winter sorghum yields during post-rainy season in medium to deep vertisols of northern dry zone of Karnataka State in South India.

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